AI-powered HDI systems are also evolving toward predictive and adaptive control, where drones can interpret operator intent and offer assistance proactively. Through machine learning, drones can learn user preferences and adjust their responses accordingly, improving overall mission efficiency and safety. Additionally, haptic feedback systems are being integrated into control devices to convey tactile information such as resistance, vibration, or force cues, enhancing situational awareness and precision.

Ultimately, developing intuitive HDI systems reduces operational workload, minimizes errors, and allows drones to act as collaborative partners rather than passive tools. This shift from control to cooperation is a defining feature of next-generation drone-human relationships.

7.10.2 Remote control challenges in autonomous and semi-autonomous drones

Despite rapid progress in autonomy, many drones still depend on varying levels of human supervision and remote control. Managing this balance between autonomy and oversight introduces several challenges related to communication latency, signal interference, situational awareness, and control reliability.

In semi-autonomous operations, the human operator must monitor the drone’s behavior and intervene when necessary. However, as drones operate farther from visual range such as in beyond visual line of sight (BVLOS) missions, the operator’s ability to perceive environmental hazards decreases. Latency in communication networks, especially when using satellite or long-distance radio links, can delay commands and sensor feedback, potentially leading to collisions or mission failure.

Furthermore, interface design complexity can hinder effective control during high-stress or time-sensitive operations. Overly detailed dashboards